

# Drakens Energy 360

An end-to-end synthetic data engineering platform for a downstream energy business

Report generated 2026-09-09 · Warehouse: drakens360\_test\_full.duckdb

**Independent synthetic portfolio project.** Drakens Energy is a fictional company invented for this project. It does not exist. Every site, customer, employee, supplier, asset, coordinate, price, volume, margin and incident in this report is randomly generated. Site coordinates are South African town centroids plus random jitter and do not represent any real service station.

## Summary

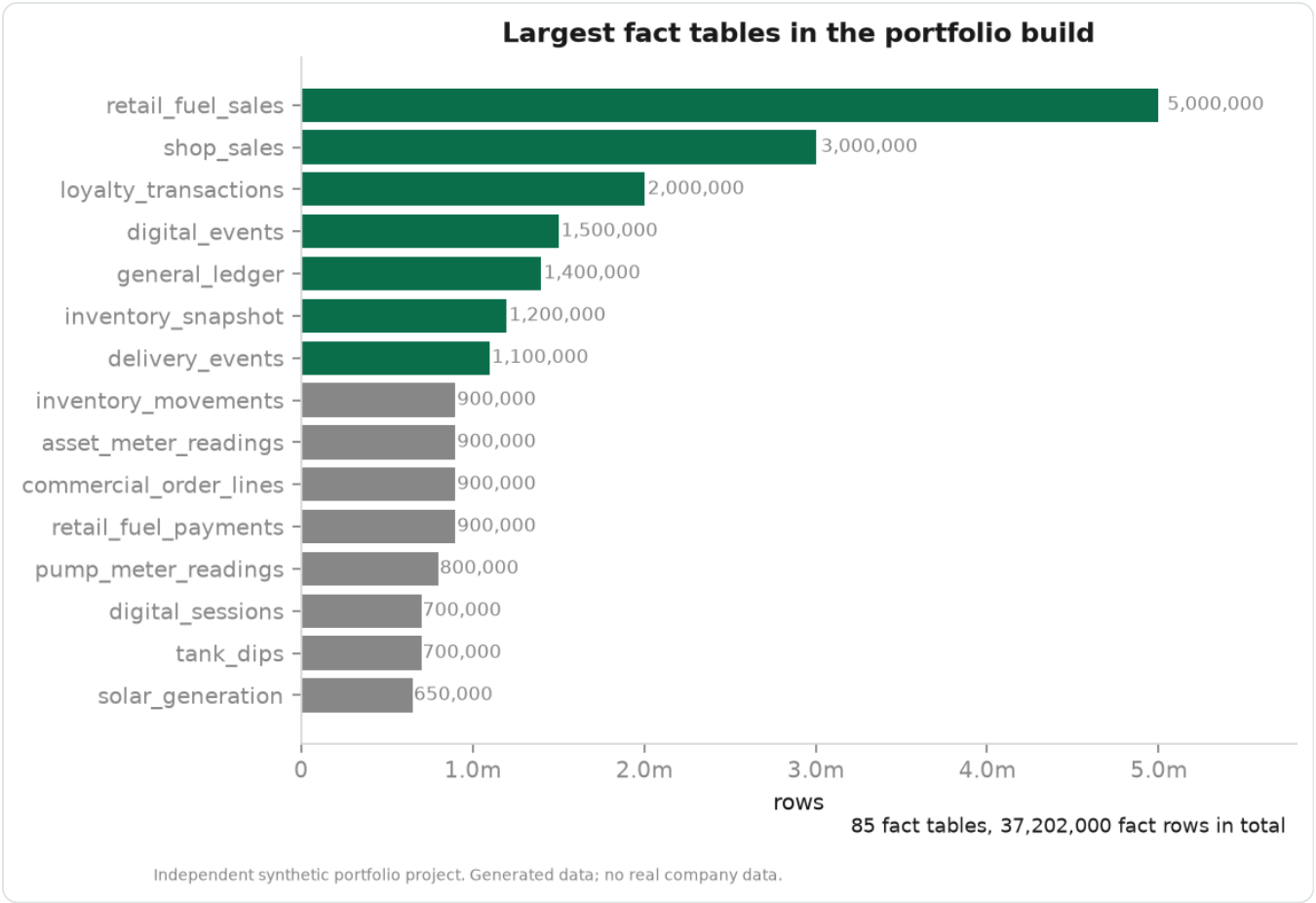
|             |            |                  |
|-------------|------------|------------------|
| FACT TABLES | FACT ROWS  | DEFECTS INJECTED |
| 85          | 37,202,000 | 62,924,526       |
| PROVINCES   |            |                  |
| 9           |            |                  |

The platform covers retail forecourt, commercial B2B, supply and distribution, LPG, lubricants, aviation, marine, loyalty, digital, EV and solar, asset maintenance, HSSEQ and finance — a downstream network of 587 synthetic service stations across 26 African markets, of which 250 are in South Africa spanning all nine provinces. It is built to demonstrate the parts of data engineering that are hard, not the parts that demo well: a deliberately dirty landing zone, a cleansing layer whose output is measured rather than asserted, and models reported against baselines they have to beat.

## Build

| Table                  | Rows      |
|------------------------|-----------|
| fact_retail_fuel_sales | 5,000,000 |
| fact_shop_sales        | 3,000,000 |

| Table                        | Rows      |
|------------------------------|-----------|
| fact_loyalty_transactions    | 2,000,000 |
| fact_digital_events          | 1,500,000 |
| fact_general_ledger          | 1,400,000 |
| fact_inventory_snapshot      | 1,200,000 |
| fact_delivery_events         | 1,100,000 |
| fact_inventory_movements     | 900,000   |
| fact_asset_meter_readings    | 900,000   |
| fact_commercial_order_lines  | 900,000   |
| fact_retail_fuel_payments    | 900,000   |
| fact_pump_meter_readings     | 800,000   |
| fact_digital_sessions        | 700,000   |
| fact_tank_dips               | 700,000   |
| fact_solar_generation        | 650,000   |
| fact_customer_invoices       | 620,000   |
| fact_site_energy_consumption | 600,000   |
| fact_customer_receipts       | 560,000   |
| fact_commercial_orders       | 550,000   |
| fact_mobile_payments         | 520,000   |
| fact_site_daily_operations   | 500,000   |
| fact_loyalty_points_balance  | 480,000   |
| fact_finance_revenue_cost    | 420,000   |
| fact_forecourt_shift         | 420,000   |
| fact_lpg_sales               | 400,000   |



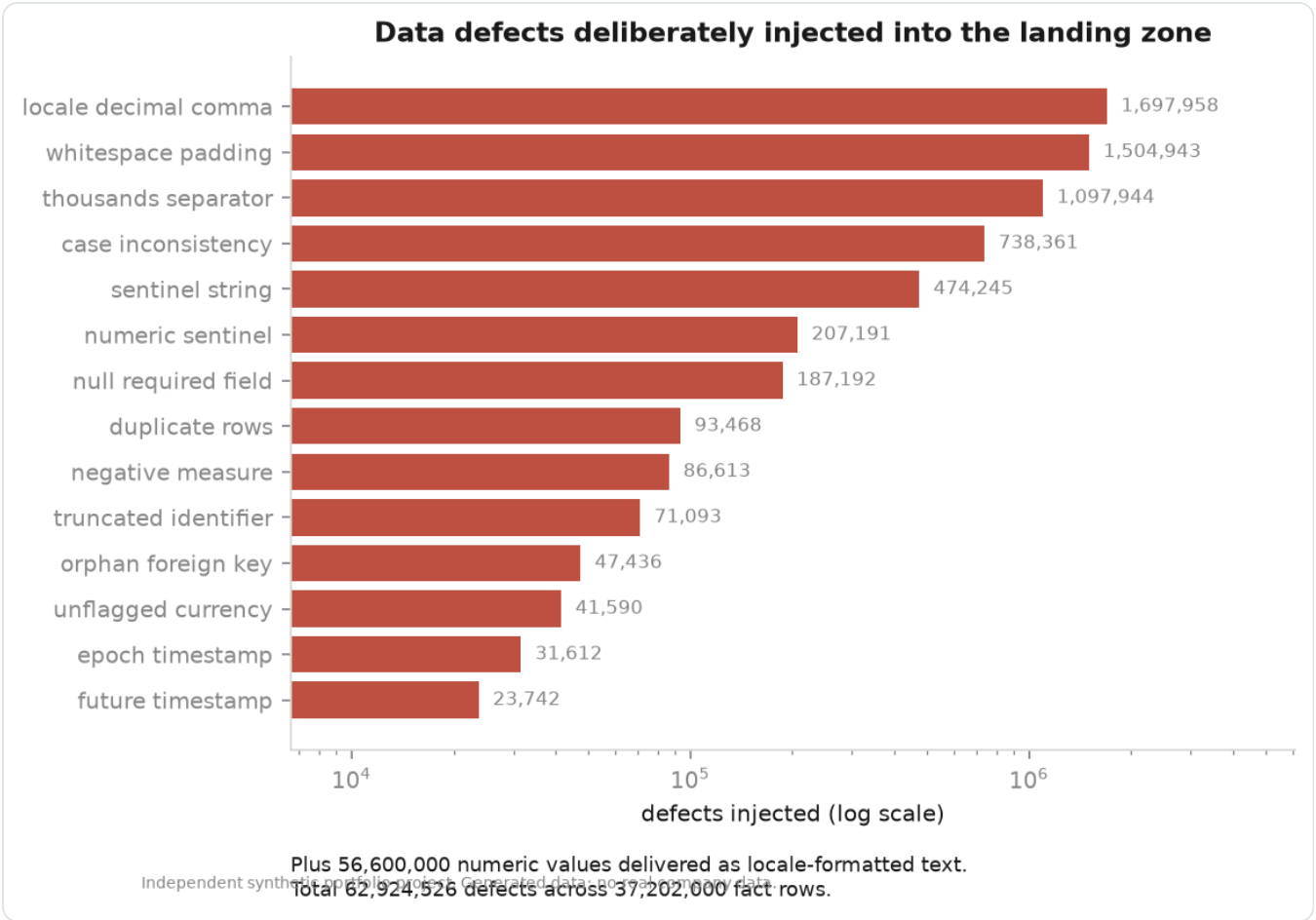
Largest fact tables in the portfolio build.

## Data quality

The landing zone is dirty on purpose. A portfolio project built on perfect data demonstrates nothing, because the difficult part of the job is deciding what to do with the row where litres reads "1 234,5", the site code has a trailing space, and the same transaction arrived twice because the point-of-sale replayed its batch. Every defect is counted as it is injected, which turns cleansing from an assertion into a measurement.

| Defect type               | Defects    |
|---------------------------|------------|
| numeric delivered as text | 56,600,000 |
| locale decimal comma      | 1,697,958  |
| whitespace padding        | 1,504,943  |
| thousands separator       | 1,097,944  |

| Defect type            | Defects |
|------------------------|---------|
| case inconsistency     | 738,361 |
| sentinel string        | 474,245 |
| numeric sentinel       | 207,191 |
| null required field    | 187,192 |
| duplicate rows         | 93,468  |
| negative measure       | 86,613  |
| truncated identifier   | 71,093  |
| orphan foreign key     | 47,436  |
| unflagged currency     | 41,590  |
| epoch timestamp        | 31,612  |
| future timestamp       | 23,742  |
| derived value mismatch | 12,256  |
| unit of measure error  | 7,233   |
| encoding mojibake      | 1,649   |



Defects deliberately injected into the landing zone.

### Cleansing outcome

| Table                       | Raw rows | Cleansed | Quarantined | Duplicates removed |
|-----------------------------|----------|----------|-------------|--------------------|
| fact_retail_fuel_sales      | 200,000  | 194,933  | 4,572       | 495                |
| fact_shop_sales             | 120,000  | 116,702  | 2,992       | 306                |
| fact_general_ledger         | 84,000   | 83,256   | 540         | 204                |
| fact_loyalty_transactions   | 80,000   | 78,944   | 857         | 199                |
| fact_delivery_events        | 66,000   | 65,557   | 283         | 160                |
| fact_digital_events         | 60,000   | 59,593   | 236         | 171                |
| fact_retail_fuel_payments   | 54,000   | 53,382   | 489         | 129                |
| fact_commercial_order_lines | 54,000   | 53,145   | 728         | 127                |
| fact_asset_meter_readings   | 54,000   | 53,453   | 397         | 150                |

| Table                        | Raw rows | Cleansed | Quarantined | Duplicates removed |
|------------------------------|----------|----------|-------------|--------------------|
| fact_inventory_movements     | 54,000   | 53,201   | 656         | 143                |
| fact_inventory_snapshot      | 50,000   | 49,300   | 583         | 117                |
| fact_pump_meter_readings     | 48,000   | 47,383   | 502         | 115                |
| fact_tank_dips               | 42,000   | 41,351   | 543         | 106                |
| fact_digital_sessions        | 42,000   | 41,745   | 162         | 93                 |
| fact_customer_invoices       | 37,200   | 36,733   | 374         | 93                 |
| fact_site_energy_consumption | 36,000   | 35,623   | 305         | 72                 |
| fact_customer_receipts       | 33,600   | 33,205   | 301         | 94                 |
| fact_mobile_payments         | 31,200   | 30,842   | 284         | 74                 |
| fact_site_daily_operations   | 30,000   | 29,556   | 365         | 79                 |
| dim_loyalty_member           | 30,000   | 30,000   | 0           | 0                  |
| fact_loyalty_points_balance  | 28,800   | 28,491   | 232         | 77                 |
| fact_solar_generation        | 26,000   | 25,704   | 228         | 68                 |
| fact_finance_revenue_cost    | 25,200   | 22,476   | 2,662       | 62                 |
| fact_forecourt_shift         | 25,200   | 24,915   | 217         | 68                 |
| dim_digital_user             | 25,000   | 25,000   | 0           | 0                  |
| fact_commercial_orders       | 25,000   | 24,417   | 528         | 55                 |
| fact_ev_charger_status       | 24,000   | 23,737   | 199         | 64                 |
| fact_grid_import_export      | 22,800   | 22,586   | 152         | 62                 |
| fact_commercial_deliveries   | 22,800   | 22,447   | 293         | 60                 |
| fact_accounts_receivable     | 22,800   | 22,508   | 232         | 60                 |
| fact_daily_executive_kpi     | 22,800   | 22,385   | 357         | 58                 |
| fact_vehicle_trips           | 20,400   | 20,247   | 100         | 53                 |
| fact_opex                    | 19,200   | 18,995   | 164         | 41                 |

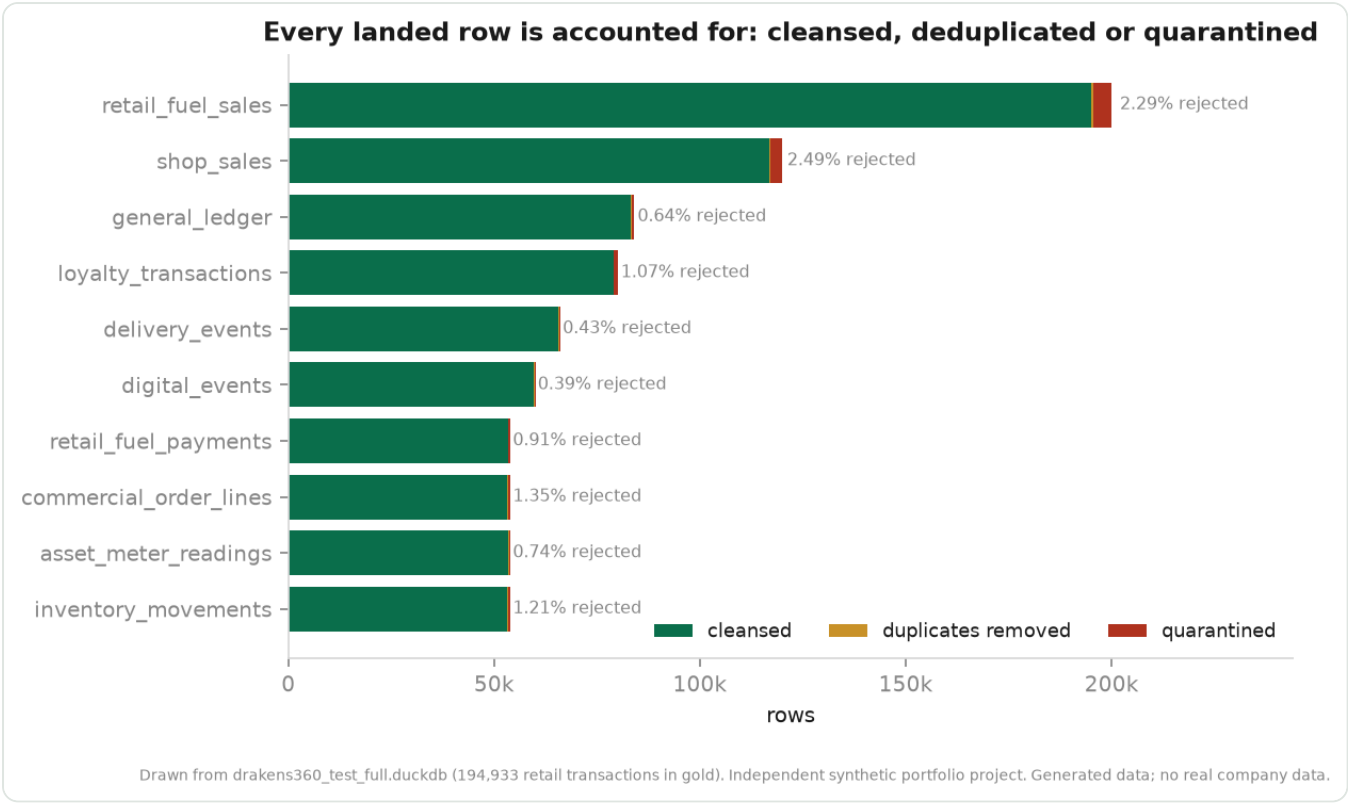
| Table                         | Raw rows | Cleansed | Quarantined | Duplicates removed |
|-------------------------------|----------|----------|-------------|--------------------|
| fact_customer_credit_exposure | 18,000   | 17,735   | 234         | 31                 |
| fact_accounts_payable         | 18,000   | 17,848   | 102         | 50                 |
| fact_forecast                 | 18,000   | 17,739   | 222         | 39                 |
| fact_lpg_cylinder_movements   | 18,000   | 17,814   | 143         | 43                 |
| fact_lpg_sales                | 16,000   | 15,627   | 331         | 42                 |
| fact_price_changes            | 15,600   | 15,353   | 204         | 43                 |
| fact_budget                   | 15,600   | 15,479   | 87          | 34                 |
| fact_replenishment_orders     | 15,600   | 15,362   | 192         | 46                 |
| fact_campaign_responses       | 15,600   | 15,470   | 92          | 38                 |
| fact_training_compliance      | 15,600   | 15,442   | 123         | 35                 |
| fact_deliveries               | 15,000   | 14,796   | 175         | 29                 |
| fact_budget_vs_actual         | 14,400   | 14,262   | 102         | 36                 |
| fact_lubricant_sales          | 14,000   | 13,687   | 281         | 32                 |
| fact_terminal_issues          | 12,600   | 12,463   | 95          | 42                 |
| fact_asset_inspections        | 12,000   | 11,879   | 93          | 28                 |
| fact_spare_parts_usage        | 11,400   | 11,292   | 83          | 25                 |
| fact_procurement_orders       | 11,400   | 11,268   | 110         | 22                 |
| fact_safety_observations      | 10,800   | 10,715   | 55          | 30                 |
| fact_fleet_costs              | 10,800   | 10,704   | 59          | 37                 |
| fact_terminal_throughput      | 10,800   | 10,686   | 93          | 21                 |
| fact_procurement_receipts     | 10,000   | 9,858    | 109         | 33                 |
| fact_energy_costs             | 9,600    | 9,470    | 111         | 19                 |
| fact_lubricant_orders         | 9,000    | 8,882    | 94          | 24                 |
| fact_route_performance        | 9,000    | 8,926    | 56          | 18                 |

| Table                             | Raw rows | Cleansed | Quarantined | Duplicates removed |
|-----------------------------------|----------|----------|-------------|--------------------|
| fact_permit_to_work               | 9,000    | 8,910    | 73          | 17                 |
| bridge_customer_contract          | 8,947    | 6,000    | 0           | 2,947              |
| fact_terminal_receipts            | 8,400    | 8,314    | 64          | 22                 |
| fact_aircraft_uplifts             | 7,800    | 7,729    | 49          | 22                 |
| fact_lpg_inventory_snapshot       | 7,200    | 7,126    | 60          | 14                 |
| fact_contract_pricing             | 7,200    | 7,123    | 62          | 15                 |
| fact_cashflow                     | 7,200    | 7,125    | 61          | 14                 |
| fact_asset_downtime               | 7,200    | 7,117    | 69          | 14                 |
| fact_inventory_adjustments        | 7,200    | 7,110    | 74          | 16                 |
| fact_lubricant_inventory_snapshot | 6,600    | 6,513    | 75          | 12                 |
| dim_customer                      | 6,000    | 5,852    | 148         | 0                  |
| bridge_customer_hierarchy         | 6,000    | 300      | 0           | 5,700              |
| fact_ev_charging_sessions         | 6,000    | 5,886    | 102         | 12                 |
| fact_working_capital              | 5,400    | 5,337    | 44          | 19                 |
| fact_vehicle_safety_events        | 5,400    | 5,352    | 34          | 14                 |
| fact_contract_volume_commitments  | 5,400    | 5,337    | 50          | 13                 |
| fact_aviation_sales               | 5,000    | 4,897    | 88          | 15                 |
| dim_asset                         | 5,000    | 5,000    | 0           | 0                  |
| bridge_asset_site                 | 5,000    | 5,000    | 0           | 0                  |
| bridge_employee_site              | 4,256    | 3,000    | 0           | 1,256              |
| fact_customer_service_cases       | 4,200    | 4,154    | 38          | 8                  |
| fact_asset_failures               | 4,200    | 4,142    | 46          | 12                 |
| fact_lpg_bulk_deliveries          | 4,200    | 4,155    | 33          | 12                 |
| fact_marine_sales                 | 4,000    | 3,921    | 71          | 8                  |



| Table                            | Raw rows | Cleansed | Quarantined | Duplicates removed |
|----------------------------------|----------|----------|-------------|--------------------|
| dim_pump                         | 4,000    | 4,000    | 0           | 0                  |
| fact_maintenance_work_orders     | 4,000    | 3,958    | 38          | 4                  |
| fact_stock_losses                | 3,600    | 3,550    | 35          | 15                 |
| fact_shop_returns                | 3,600    | 3,548    | 41          | 11                 |
| fact_bunker_deliveries           | 3,600    | 3,571    | 18          | 11                 |
| bridge_site_product              | 3,536    | 600      | 0           | 2,936              |
| dim_contract                     | 3,500    | 3,500    | 0           | 0                  |
| bridge_customer_site             | 3,172    | 1,500    | 0           | 1,672              |
| dim_employee                     | 3,000    | 3,000    | 0           | 0                  |
| fact_near_misses                 | 2,760    | 2,729    | 24          | 7                  |
| dim_tank                         | 2,500    | 2,500    | 0           | 0                  |
| fact_capex                       | 2,400    | 2,369    | 26          | 5                  |
| fact_promotions                  | 2,400    | 397      | 223         | 1,780              |
| fact_aviation_inventory_snapshot | 2,400    | 2,365    | 27          | 8                  |
| bridge_route_site                | 1,869    | 600      | 0           | 1,269              |
| fact_marine_inventory_snapshot   | 1,800    | 1,776    | 16          | 8                  |
| fact_environmental_events        | 1,320    | 1,296    | 18          | 6                  |
| dim_equipment                    | 1,250    | 1,250    | 0           | 0                  |
| bridge_supplier_product          | 1,206    | 400      | 0           | 806                |
| dim_driver                       | 1,100    | 1,100    | 0           | 0                  |
| bridge_site_promotion            | 1,018    | 600      | 0           | 418                |
| dim_vehicle                      | 900      | 900      | 0           | 0                  |
| fact_hsseq_incidents             | 800      | 795      | 4           | 1                  |
| bridge_site_hierarchy            | 600      | 586      | 14          | 0                  |

| Table                    | Raw rows | Cleansed | Quarantined | Duplicates removed |
|--------------------------|----------|----------|-------------|--------------------|
| dim_route                | 600      | 600      | 0           | 0                  |
| dim_site                 | 600      | 586      | 14          | 0                  |
| fact_import_cargoes      | 540      | 538      | 2           | 0                  |
| dim_ev_charger           | 400      | 400      | 0           | 0                  |
| dim_supplier             | 400      | 385      | 15          | 0                  |
| dim_solar_asset          | 300      | 300      | 0           | 0                  |
| dim_city                 | 88       | 88       | 0           | 0                  |
| dim_terminal             | 60       | 60       | 0           | 0                  |
| dim_promotion            | 44       | 44       | 0           | 0                  |
| dim_depot                | 40       | 40       | 0           | 0                  |
| dim_campaign             | 38       | 38       | 0           | 0                  |
| dim_product              | 31       | 31       | 0           | 0                  |
| bridge_product_hierarchy | 31       | 31       | 0           | 0                  |
| dim_price_zone           | 30       | 30       | 0           | 0                  |
| dim_country              | 29       | 28       | 1           | 0                  |
| dim_currency             | 25       | 25       | 0           | 0                  |
| dim_carrier              | 22       | 22       | 0           | 0                  |
| dim_warehouse            | 20       | 20       | 0           | 0                  |
| dim_airport              | 12       | 12       | 0           | 0                  |
| dim_legal_entity         | 12       | 12       | 0           | 0                  |
| dim_province             | 9        | 9        | 0           | 0                  |
| dim_port                 | 8        | 8        | 0           | 0                  |
| dim_lpg_cylinder_type    | 4        | 4        | 0           | 0                  |



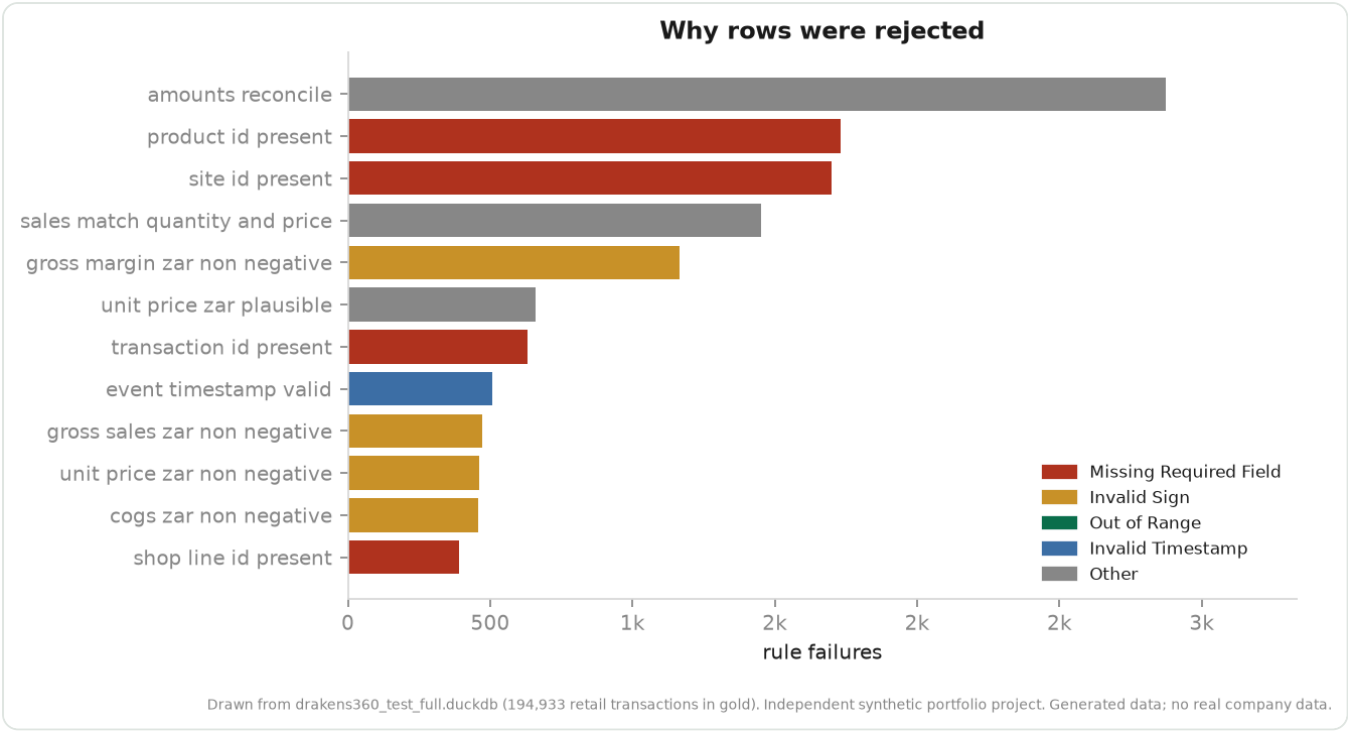
Every landed row is accounted for as cleansed, deduplicated or quarantined.

Why rows were rejected

| Rule that rejected the row     | Category               | Likely owner     | Rows   |
|--------------------------------|------------------------|------------------|--------|
| amounts_reconcile              | Other                  | Data Platform    | 2,873  |
| product_id_present             | Missing Required Field | Source System    | 1,731  |
| site_id_present                | Missing Required Field | Source System    | 1,699  |
| sales_match_quantity_and_price | Other                  | Data Platform    | 1,452  |
| gross_margin_zar_non_negative  | Invalid Sign           | Business Process | 1,166  |
| unit_price_zar_plausible       | Other                  | Data Platform    | 660.00 |
| transaction_id_present         | Missing Required Field | Source System    | 630.00 |
| event_timestamp_valid          | Invalid Timestamp      | Source System    | 506.00 |
| gross_sales_zar_non_negative   | Invalid Sign           | Business Process | 473.00 |
| unit_price_zar_non_negative    | Invalid Sign           | Business Process | 460.00 |
| cogs_zar_non_negative          | Invalid Sign           | Business Process | 458.00 |

| Rule that rejected the row        | Category               | Likely owner     | Rows   |
|-----------------------------------|------------------------|------------------|--------|
| shop_line_id_present              | Missing Required Field | Source System    | 391.00 |
| journal_line_id_present           | Missing Required Field | Source System    | 311.00 |
| loyalty_transaction_id_present    | Missing Required Field | Source System    | 284.00 |
| litres_non_negative               | Invalid Sign           | Business Process | 253.00 |
| digital_event_id_present          | Missing Required Field | Source System    | 197.00 |
| reading_id_present                | Missing Required Field | Source System    | 176.00 |
| litres_within_range               | Out of Range           | Data Entry       | 161.00 |
| snapshot_id_present               | Missing Required Field | Source System    | 160.00 |
| fill_pct_in_range                 | Other                  | Data Platform    | 143.00 |
| tank_dip_id_present               | Missing Required Field | Source System    | 125.00 |
| points_value_zar_non_negative     | Invalid Sign           | Business Process | 117.00 |
| debit_amount_zar_non_negative     | Invalid Sign           | Business Process | 93.00  |
| solar_reading_id_present          | Missing Required Field | Source System    | 91.00  |
| order_id_present                  | Missing Required Field | Source System    | 91.00  |
| qualifying_spend_zar_non_negative | Invalid Sign           | Business Process | 90.00  |
| credit_amount_zar_non_negative    | Invalid Sign           | Business Process | 88.00  |
| volume_litres_non_negative        | Invalid Sign           | Business Process | 88.00  |
| customer_id_present               | Missing Required Field | Source System    | 88.00  |
| capacity_litres_non_negative      | Invalid Sign           | Business Process | 83.00  |
| stock_on_hand_litres_non_negative | Invalid Sign           | Business Process | 69.00  |
| volume_litres_within_range        | Out of Range           | Data Entry       | 59.00  |
| delivery_id_present               | Missing Required Field | Source System    | 56.00  |
| revenue_zar_non_negative          | Invalid Sign           | Business Process | 43.00  |
| energy_kwh_non_negative           | Invalid Sign           | Business Process | 39.00  |

| Rule that rejected the row      | Category               | Likely owner     | Rows  |
|---------------------------------|------------------------|------------------|-------|
| ev_session_id_present           | Missing Required Field | Source System    | 33.00 |
| energy_kwh_within_range         | Out of Range           | Data Entry       | 32.00 |
| fill_rate_pct_in_range          | Other                  | Data Platform    | 28.00 |
| list_price_zar_non_negative     | Invalid Sign           | Business Process | 27.00 |
| delivered_litres_non_negative   | Invalid Sign           | Business Process | 19.00 |
| delivered_litres_within_range   | Out of Range           | Data Entry       | 13.00 |
| work_order_id_present           | Missing Required Field | Source System    | 8.00  |
| grid_cost_zar_non_negative      | Invalid Sign           | Business Process | 8.00  |
| rated_power_kw_non_negative     | Invalid Sign           | Business Process | 7.00  |
| labour_cost_zar_non_negative    | Invalid Sign           | Business Process | 7.00  |
| total_cost_zar_non_negative     | Invalid Sign           | Business Process | 7.00  |
| average_power_kw_non_negative   | Invalid Sign           | Business Process | 3.00  |
| parts_cost_zar_non_negative     | Invalid Sign           | Business Process | 3.00  |
| incident_id_present             | Missing Required Field | Source System    | 2.00  |
| estimated_cost_zar_non_negative | Invalid Sign           | Business Process | 1.00  |



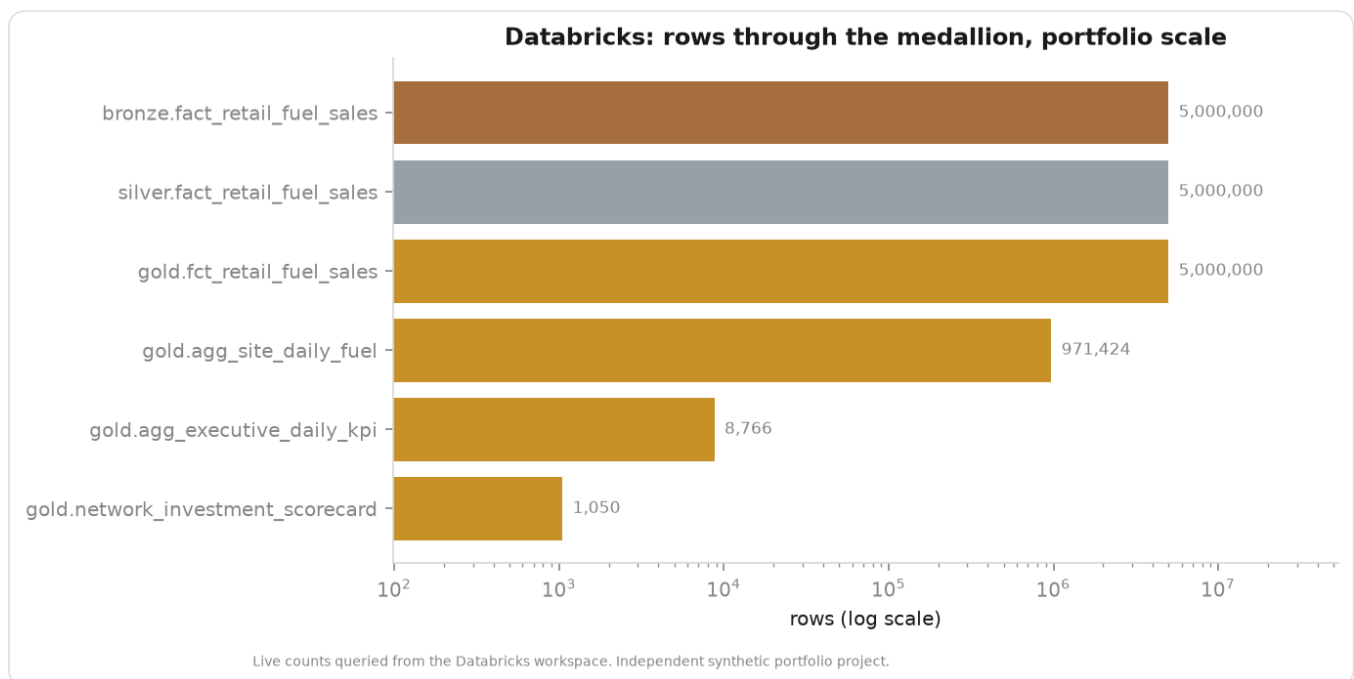
Rejection reasons. The operational response differs by rule.

## Warehouse

| Control                           | What it checks                                                 | Owner      | Variance % | Result |
|-----------------------------------|----------------------------------------------------------------|------------|------------|--------|
| DELIVERIES_HAVE_KNOWN_DESTINATION | Every delivery resolves to a site in the conformed dimension   | Supply     | 0.00       | PASS   |
| GL_DEBITS_EQUAL_CREDITS           | Posted journal debits equal credits across the ledger          | Finance    | 0.00       | PASS   |
| ORDERS_HAVE_KNOWN_CUSTOMER        | Every commercial order resolves to a customer in the dimension | Commercial | 0.00       | PASS   |
| RETAIL_MARGIN_RECONCILES          | Retail fuel gross margin equals revenue less cost of sales     | Finance    | 0.00       | PASS   |

| Control                         | What it checks                                         | Owner  | Variance % | Result |
|---------------------------------|--------------------------------------------------------|--------|------------|--------|
| RETAIL_VOLUME_TIES_TO_DAILY_AGG | Transaction-level litres tie to the site-day aggregate | Supply | 0.00       | PASS   |

A reconciliation control evaluates a published number as data rather than only as a test, so a control that fails names the figure that is wrong and by how much. Controls that fail are shown as failures; a report that quietly omits them would defeat the purpose of having them.

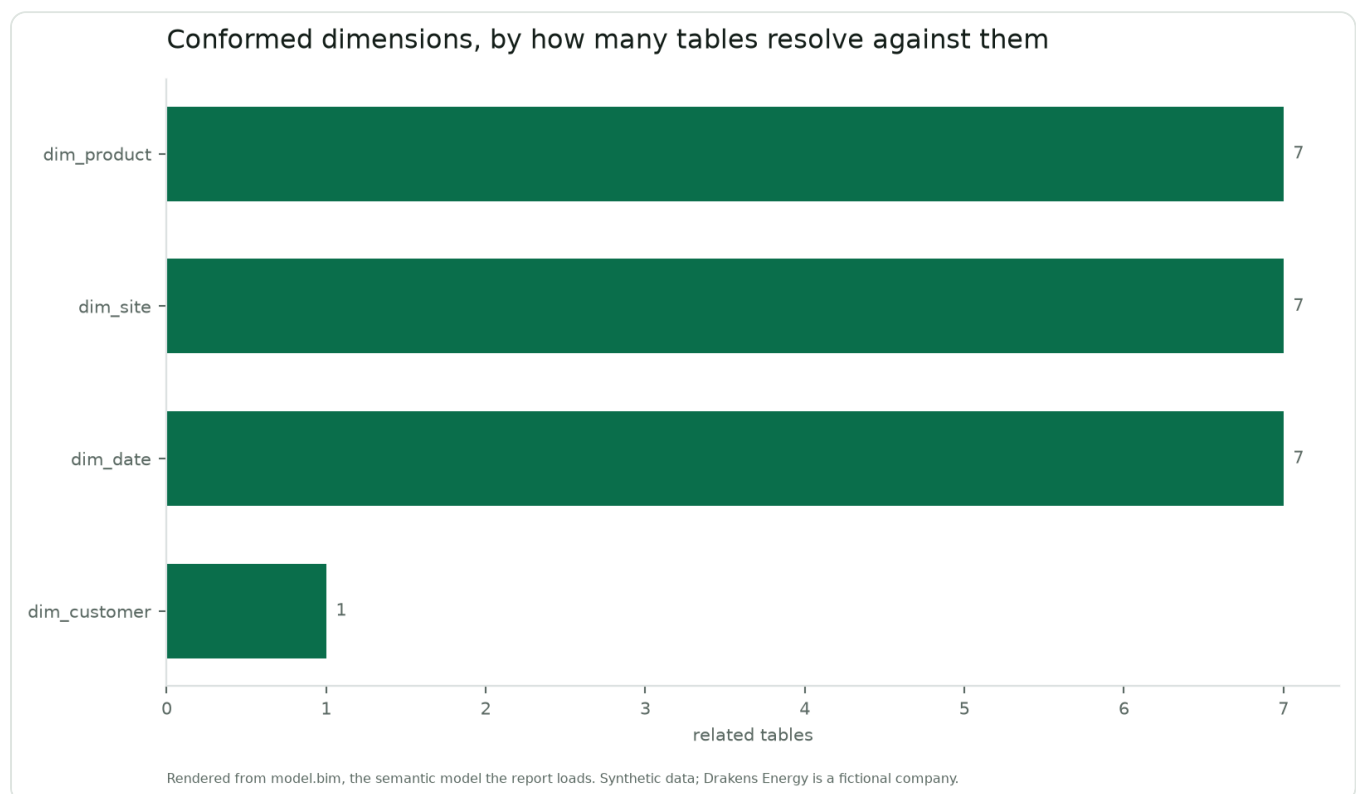


Live row counts through bronze, silver and gold on Databricks at portfolio scale.

## The data model

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Every diagram below is rendered from `model1.bim`, the semantic model the Power BI report loads — not drawn by hand and not transcribed from a design document. Each box, edge and cardinality is therefore a statement about the artifact that was actually built, and a relationship added tomorrow appears in tomorrow's diagram. Where a fact has no modelled relationship the diagram says so, because that absence is a finding rather than a layout accident.

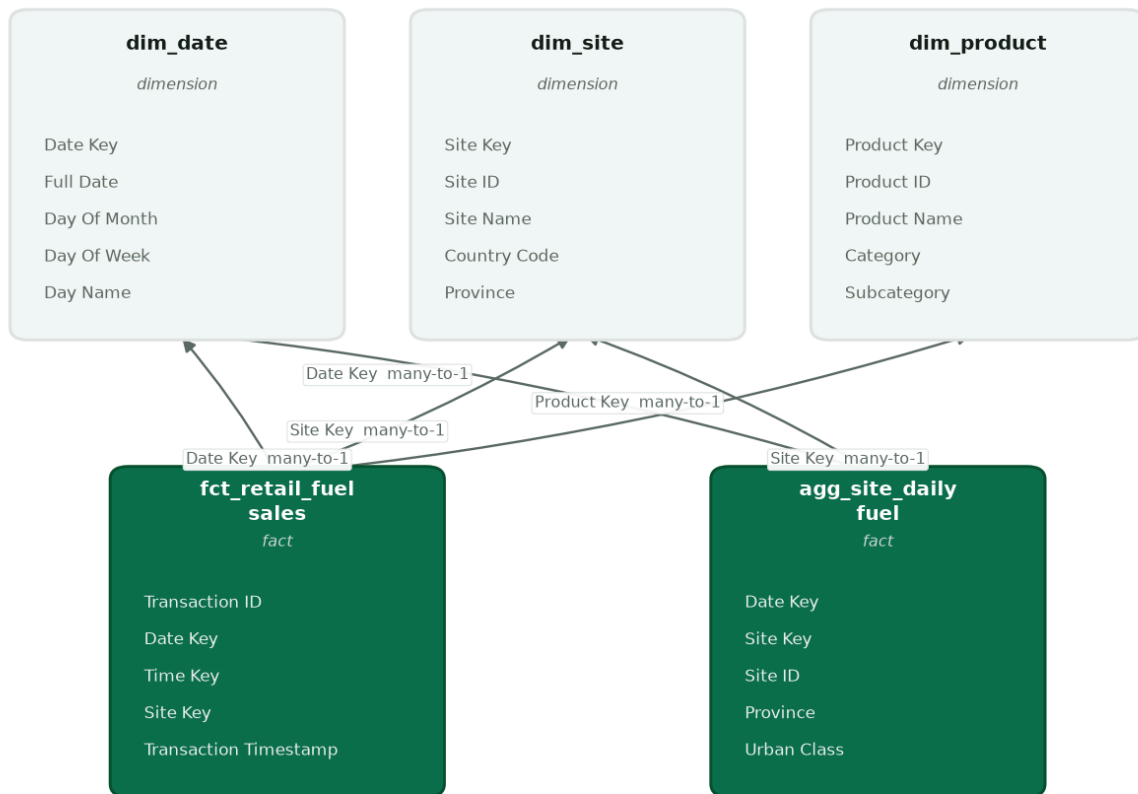


Conformed dimensions ranked by how many tables resolve against them. A dimension that only one fact uses is not conformed; it is a lookup table with ambitions.



## Retail fuel

2 fact table(s), 3 related table(s)

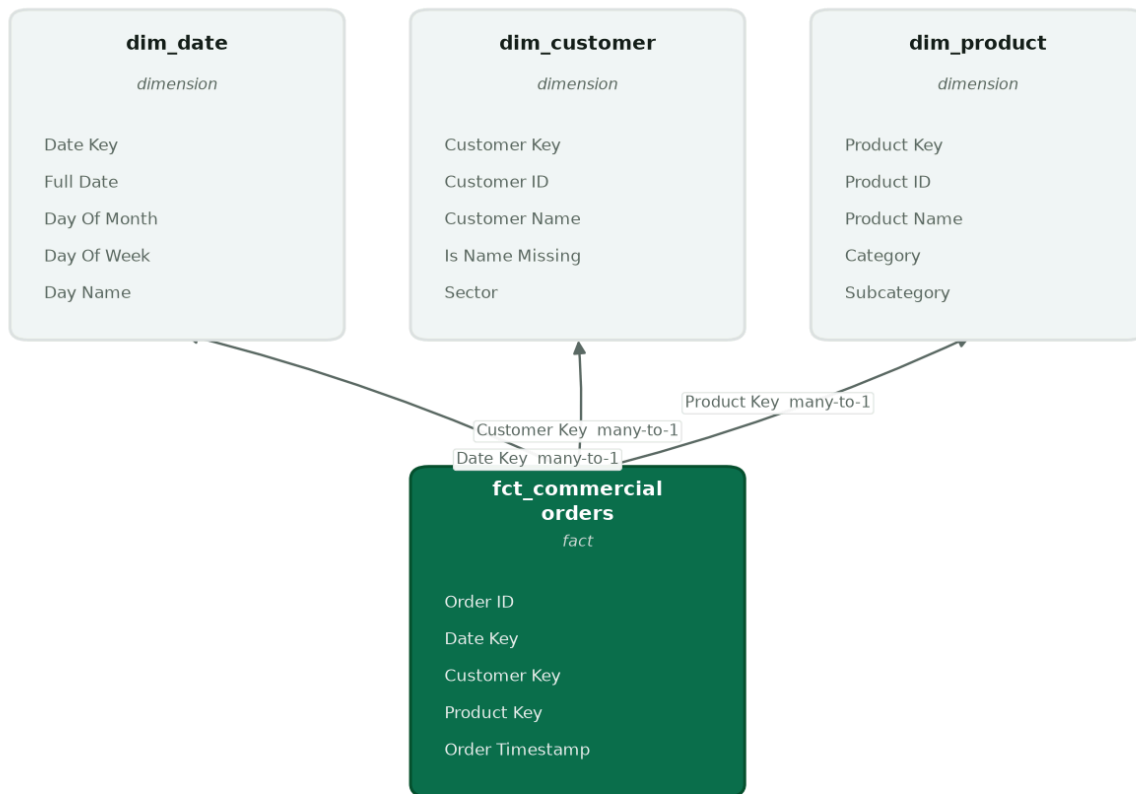


Rendered from model.bim, the semantic model the report loads. Synthetic data; Drakens Energy is a fictional company.

Retail fuel. The transaction fact and the site-day aggregate resolve against the same three dimensions, which is what makes the aggregate reconcilable against the detail rather than merely similar to it.

## Commercial B2B

1 fact table(s), 3 related table(s)



Rendered from model.bim, the semantic model the report loads. Synthetic data; Drakens Energy is a fictional company.

Commercial B2B. The relationship to dim\_product was missing until recently, which meant lubricants — the largest line in the business at R20.3bn — could not be sliced by product at all.

## Supply and distribution

1 fact table(s), 2 related table(s)

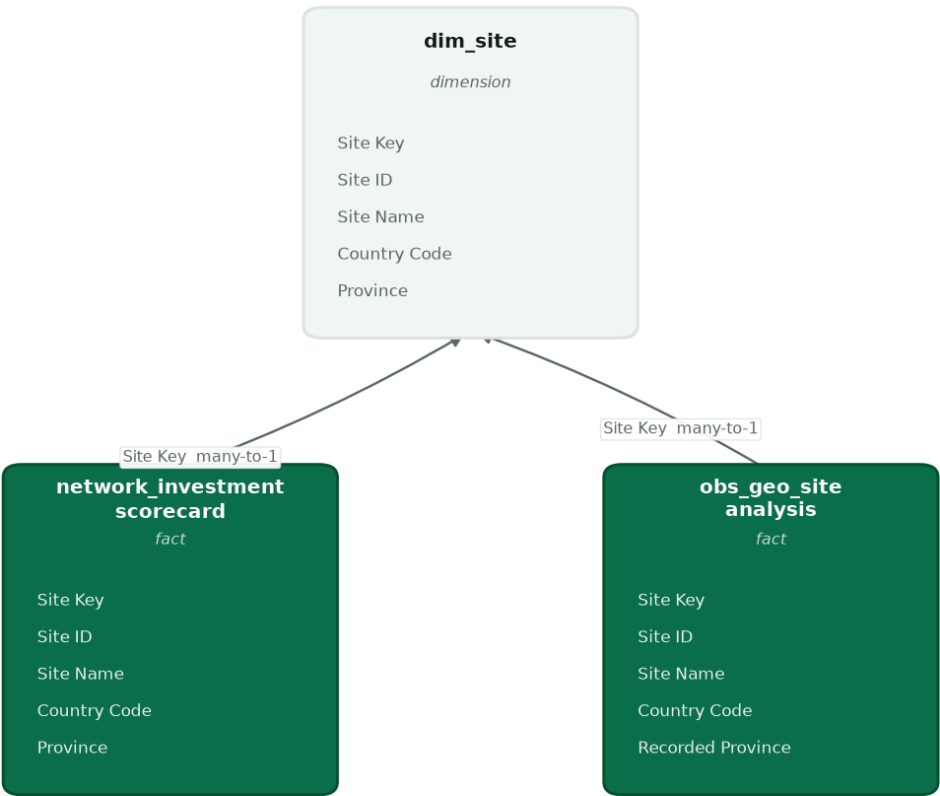


Rendered from model.bim, the semantic model the report loads. Synthetic data; Drakens Energy is a fictional company.

Supply and distribution. Deliveries resolve to a destination site, so a delivery to an unrecognised site lands on the unknown member rather than vanishing from the count.

# Network investment and geography

2 fact table(s), 1 related table(s)

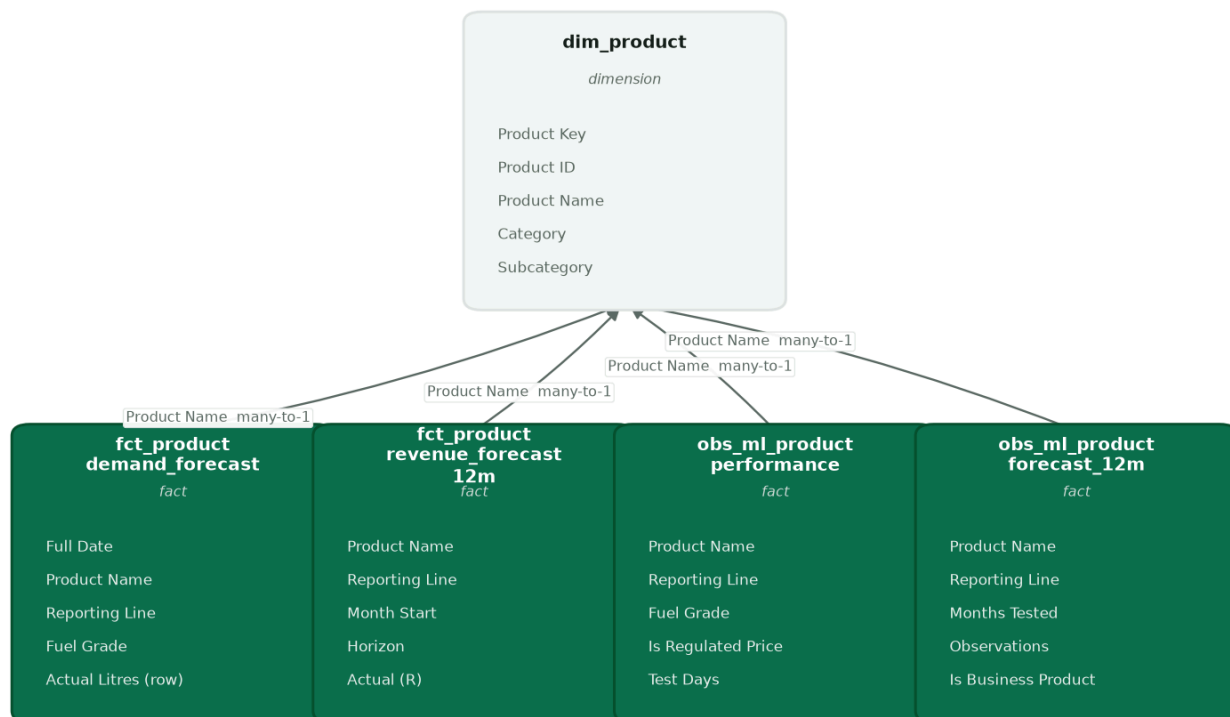


Rendered from model.bim, the semantic model the report loads. Synthetic data; Drakens Energy is a fictional company.

Network investment and the spatial analysis, both keyed on the site dimension.

## Forecasting and model results

4 fact table(s), 1 related table(s)



Rendered from model.bim, the semantic model the report loads. Synthetic data; Drakens Energy is a fictional company.

Forecasting and model results. These tables key on product name rather than a surrogate, because they are produced by the ML layer rather than by dbt.

## What the diagrams do not cover

| Reporting line | Products | Revenue (R)    | Status       | Note |
|----------------|----------|----------------|--------------|------|
| Lubricant      | 7        | 20,256,345,787 | Reportable   | nan  |
| Road Fuel      | 4        | 3,166,357,716  | Reportable   | nan  |
| Unknown        | 1        | 34,600,027     | Reportable   | nan  |
| Other Fuel     | 1        | 4,104,049      | Reportable   | nan  |
| Energy         | 3        | 1,665,959      | Reportable   | nan  |
| Aviation       | 2        | 0.00           | No gold fact | nan  |
| Convenience    | 8        | 0.00           | No gold fact | nan  |
| LPG            | 4        | 0.00           | No gold fact | nan  |
| Marine         | 2        | 0.00           | No gold fact | nan  |

Five of the nine reporting lines cannot be charted, and for three different reasons. Four have no gold fact at all. Energy has one — `fct_ev_charging_sessions`, carrying R1.67m of revenue and 232,029 kWh — and no `product_key`, so it cannot join the product dimension. Reading a blank bar as “no data” collapses three different problems into the wrong one.

# Implementation walkthrough

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How the platform is built, stage by stage, with the decision at each one and the evidence it produced. Command output shown below is captured from actual runs and is labelled as command output; it is not a screenshot of a cloud console.

## Databricks implementation walkthrough

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Drakens Energy is a fictional business. This portfolio uses generated data, including jittered town-centroid coordinates; it does not represent real service stations or company records.

This walkthrough explains the implementation present in the repository, the decisions it demonstrates, and the evidence needed to validate a run. Source review is not a cloud execution result. Local DuckDB results, Spark notebook results and Power BI captures must retain their own environment, profile, date and commit context.

### 1. Business questions and data design

The platform connects fuel transactions, sites, products, customers and operational events so a reviewer can follow margin, service reliability, replenishment and investment decisions back to source records. Conformed dimensions allow the same site to be analysed across subjects without separately maintained definitions.

Start with the [implementation guide](IMPLEMENTATION\_GUIDE.md), [data dictionary](data\_dictionary.md), [subject-area ERDs](erd/README.md), and the [Gold retail ERD](erd/erd\_gold\_retail\_validated.md). State the grain before describing a metric: a transaction line, a site-day aggregate and a site scorecard are different populations.

**Evidence to show:** an architecture diagram, source inventory, a small source sample, and a model diagram with keys and fact grain. Explain the choice of transactional detail plus aggregates as a trade-off between traceability and report query cost.

### 2. Select the execution path and scale

The repository contains several paths; they are not interchangeable run records.

| Path                   | Entry point                                                          | What the source implements                                                      | Evidence required for an execution claim                                   |
|------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Local reference build  | <code>src/drakens360/build.py</code> , dbt models                    | Python-generated landing data and a local warehouse used by CI                  | Selected manifest, warehouse path, command output, commit and test results |
| Spark-native batch job | <code>scripts/deploy_databricks.py</code> , notebooks 00/02/03/05/06 | In-workspace generation followed by conformance, marts, quality and maintenance | Workspace job/run ID, task outcomes and actual output counts               |
| File ingestion         | Notebook 01                                                          | Auto Loader configuration for landed feeds                                      | Source file, checkpoint/schema location and completed ingestion evidence   |
| Streaming example      | Notebook 04                                                          | Demo rate source and a separate Kafka source branch                             | Mode, actual source schema, query progress and restart/late-event evidence |

The notebook 00 SCALES dictionary currently configures portfolio generation with 1,050 ZA sites and 4,200 total sites. The previously inspected local report warehouse contains 250 ZA sites and 587 total sites. These are different implementations and profiles. Do not replace one with the other's total or say that all code generates the same estate. Measure the particular artifact used by each figure.

### 3. Workspace and catalog preparation

`[scripts/deploy_databricks.py](../scripts/deploy_databricks.py)` selects a configured authentication profile, uses an existing SQL warehouse, creates an environment catalog and schemas, uploads notebooks, and creates or resets the named job. `--run` triggers execution; deployment alone does not demonstrate a successful run. The code selects the first returned SQL warehouse, so a reproducibility record must identify which warehouse was actually used.

`[01_catalog_and_schemas.sql](../databricks/sql/01_catalog_and_schemas.sql)` also defines managed volumes, including landing and checkpoints. Notebook upload, schema creation and volume preparation are separate concerns: the deployment helper's catalog function creates schemas, not every volume referenced by optional ingestion notebooks. Inspect required locations before attempting that path.



Record the environment, runtime/compute, dependency versions, catalog, notebook revision and selected scale. Use configuration names in screenshots; exclude secret values. Terraform and Bicep files under `infra/` describe an Azure infrastructure option, and should be labelled as provisioned only when deployment evidence exists.

**Skill demonstrated:** reproducible environment configuration, separation of environments, explicit dependencies and avoidance of credentials in source.

#### 4. Generate and inspect Bronze data

[00\_generate\_bronze\_at\_scale.py](../databricks/notebooks/00\_generate\_bronze\_at\_scale.py) uses Spark expressions and scale/seed widgets to create synthetic records in the chosen catalog. The local generator is a separate reference path. Inspect the manifest for the selected local build rather than assuming a notebook run produced that manifest.

Show a small, named input sample with examples of injected defects: numeric text, whitespace, sentinels, duplicate business keys or broken references. Explain which layer will handle each defect. Raw row counts and defect counts describe different things because one record can contain several defects.

**Capture:** parameters, completed generation output, table preview and the corresponding manifest where applicable. A generated analytical plot is useful here but does not replace execution evidence.

#### 5. Explain the optional Auto Loader path

[01\_bronze\_autoloader.py](../databricks/notebooks/01\_bronze\_autoloader.py) has a feed registry for forecourt, shop, commercial, finance and maintenance inputs. It constructs per-feed checkpoint and schema paths and configures `cloudFiles` options including `addNewColumns`, a rescued-data column and a micro-batch file limit.

Document the exact feed, input format, source path, schema location, checkpoint and trigger mode used. Show that a second ingestion of the same input does not unexpectedly increase the target and explain what happens when an upstream schema changes. Configuration alone does not prove restart behavior or uninterrupted schema evolution.

The default generated batch job does not include notebook 01. Keep it visibly separate in the workflow diagram, and do not imply it executed because the five-task batch job succeeded.

#### 6. Silver conformance and quarantine

[02\_silver\_conformance.py](../databricks/notebooks/02\_silver\_conformance.py) declares contracts per feed. For retail fuel, examples include required transaction/site/product identifiers, plausible litres and unit prices, and gross-margin reconciliation. It records rule outcomes, writes rejected records to

platform.quarantine, ranks records for deduplication using business keys and timestamps, and writes Silver output.

Use the same business key in the before/after example. Show raw values, the applied rule, accepted output or rejection reason, and the associated run. Reconcile the input population with accepted records, quarantined records and duplicate removal using the implementation's actual counting order; do not add overlapping categories as though they were mutually exclusive.

The notebook includes a dimension SCD2 merge implementation. Demonstrate history with a controlled changed attribute and the resulting current/expired records before claiming the behavior is verified. Distinguish storing dimension history from resolving a fact to the correct historical version.

**Skill demonstrated:** explicit data contracts, traceable cleaning, deterministic deduplication and explainable rejection handling.

## 7. Gold marts, keys and grain

[03\_gold\_star\_schema.py](../databricks/notebooks/03\_gold\_star\_schema.py) builds Spark-native dimensions and marts. The dbt models provide a separate path under dbt/models/marts/. Both should be explained with their real definitions rather than described as identical merely because table names resemble each other.

The dbt retail fact is one transaction line per transaction\_id. It resolves the current site/product dimension by business identifier and maps unmatched keys to the unknown member -1, retaining flags that make drift countable. The site-day aggregate groups by date\_key and site\_key. Revenue, cost, litres and margin can be summed over compatible populations; a margin percentage should be recomputed from its numerator and denominator.

Important historical limitation: the inspected dbt retail join filters dimensions to is\_current; notebook 03 also selects current site records and assigns site\_key from site\_id. Neither detail should be described as proof of an as-of historical fact-to-dimension join. An event-time interval join and controlled historical test would be required to demonstrate that claim.

See the [validated Gold retail model](erd/erd\_gold\_retail\_validated.md) for logical relationships and the distinction from source-manifest diagrams.

## 8. Orchestration and failure boundaries

The actual sequence declared by create\_job() is:

```
flowchart LR
    A[generate_bronze / notebook 00] --> B[build_silver / notebook 02]
    B --> C[build_gold / notebook 03]
```

```
C --> D[data_quality / notebook 05]
D --> E[optimize_tables / notebook 06]
```

The helper sets one concurrent run and a task timeout. The listed dependencies explain execution order; do not add a schedule, automatic retry policy, Auto Loader or streaming task to this diagram unless the job definition is changed and verified.

**Capture:** the actual job graph with run ID and final task states. Include a failed task, its diagnosis and rerun evidence where available. Distinguish a repaired local test from a successful remote workflow.

## 9. Streaming telemetry and its current boundary

[04\_streaming\_telemetry.py](../databricks/notebooks/04\_streaming\_telemetry.py) provides tank, fleet, EV and solar examples. Its demo source yields rate-stream value and timestamp columns. The production branch returns raw\_json and kafka\_ts, while the inspected downstream tank transformation still refers to value and timestamp. A payload decode/schema adapter is therefore a prerequisite to claiming this production branch works end to end.

Explain event time, configured watermarks, checkpoint paths and the intended destination using the actual code. Show query progress and a late-event/restart test only when executed. A demo generator is not evidence of sustained Event Hubs ingestion. Preserve this boundary in the ebook even when the architecture includes Event Hubs as a planned source.

## 10. Quality, freshness and governance

[05\_data\_quality\_checks.py](../databricks/notebooks/05\_data\_quality\_checks.py) produces operational results, including freshness records with a run ID and evaluation time. It distinguishes old backfill data from recent-feed SLA checks and can fail a critical check according to its widget setting. The inspected MONITORED list contains SLA constants; do not describe them as dynamically loaded reference data without a code change.

[02\_governance.sql](../databricks/sql/02\_governance.sql) contains group grants, masks and row-filter logic. The SQL file is implementation evidence; proof of effective permissions requires queries under representative identities. Show a permitted result and a restricted result without displaying credentials or unrelated personal data.

**Capture:** quality output, reconciliation differences, quarantine reasons and effective-access evidence. A green control must remain tied to the run, table and population it checked.

## 11. Performance and maintenance

[06\_optimize\_and\_vacuum.py](../databricks/notebooks/06\_optimize\_and\_vacuum.py) describes a per-table maintenance plan and captures table statistics. It exposes run\_vacuum and a retention setting, rejects retention below its 168-hour floor, and follows quality checks in the batch graph.

Present observed file count, table size, runtime and query behavior only with the relevant before/after capture and workload context. Comments reporting an earlier measurement are historical claims until their supporting artifact is identified. A change in file count alone is not a benchmark of query speed or a monetary saving.

Document the effect of retention and maintenance on recoverability. Do not execute maintenance or provision compute merely to fill a screenshot slot.

## 12. ML, decision support and report consumption

The experiments in ml/experiments/ and the MLflow export in [export\_ml\_results.py](../scripts/export\_ml\_results.py) provide model evaluation evidence. Explain target, grain, train/test split, leakage controls, metric direction and baseline for each model. AUC is higher-is-better; MAE is lower-is-better. The export's raw score-minus-baseline delta has opposite improvement directions for these metrics and should not be ranked on one mixed-unit chart.

The unsupervised anomaly experiment has a different evaluation status from the scored supervised models. Preserve weak results and missing scores. The network investment engine is a constrained optimization component; do not present its rules or allocation as a learned prediction model.

Show how exported evaluation results reach the Power BI model. Evaluation summary rows are not per-site prediction records. For actual prediction use, identify the scored entity/date grain, model/run identity and consuming visual. Power BI page captures must prove the rendered output, and paired captures should show changed metrics when a South Africa filter changes.

## 13. Reproducibility and portfolio evidence

The implementation guide owns commands and environment setup. [reconcile\_figures.py](../scripts/reconcile\_figures.py) checks document figures against source artifacts; a fresh successful result belongs beside the final generated PDF/HTML versions. Keep the workflow's Git commit and CI result separate from a Databricks run ID.

Use this coverage register when preparing final document figures:

| Evidence               | Suitable capture                                    | What it proves                                      |
|------------------------|-----------------------------------------------------|-----------------------------------------------------|
| Environment and scale  | Configuration plus run header                       | Which implementation and population were used       |
| Source and Bronze      | Actual sample and generation/ingestion result       | Input content and completed landing step            |
| Silver                 | Same-key before/after and quarantine result         | Transformation or rejection of identifiable records |
| Gold and ERD           | Schema, key tests and grain checks                  | Model structure and relationship validity           |
| Job execution          | Completed job task graph                            | Which tasks ran and their outcomes                  |
| Streaming              | Query progress and controlled late/restart examples | The specific streaming behavior exercised           |
| Quality and governance | Control results and representative access queries   | Validation and effective restrictions for that run  |
| ML                     | Experiment metrics, split details and baseline      | Evaluation rather than unsupported accuracy claims  |
| Power BI and Excel     | Actual pages, filters and formatted results         | Correct and readable consumption                    |
| Final build            | Reconciliation and completed CI result              | Traceability of the delivered version               |

Authentic UI captures remain outstanding where desktop/workspace access is unavailable. Clearly label captured command output as execution output, analytical charts as charts, and missing UI captures as missing. Number figures in document order and keep captions with images. This walkthrough is source-verified documentation; it does not certify those outstanding runtime checks.

#### COMMAND OUTPUT — SPATIAL AXIS GUARD

ST\_Distance\_Sphere axis check: Johannesburg to Cape Town

expected 1,261,576 m

measured 1,261,576 m PASS

point-in-polygon check (against Natural Earth admin-1):

Not South Africa 336

Matches 232

Outside the mapped province polygons 15

Province mismatch 2

No province recorded 1

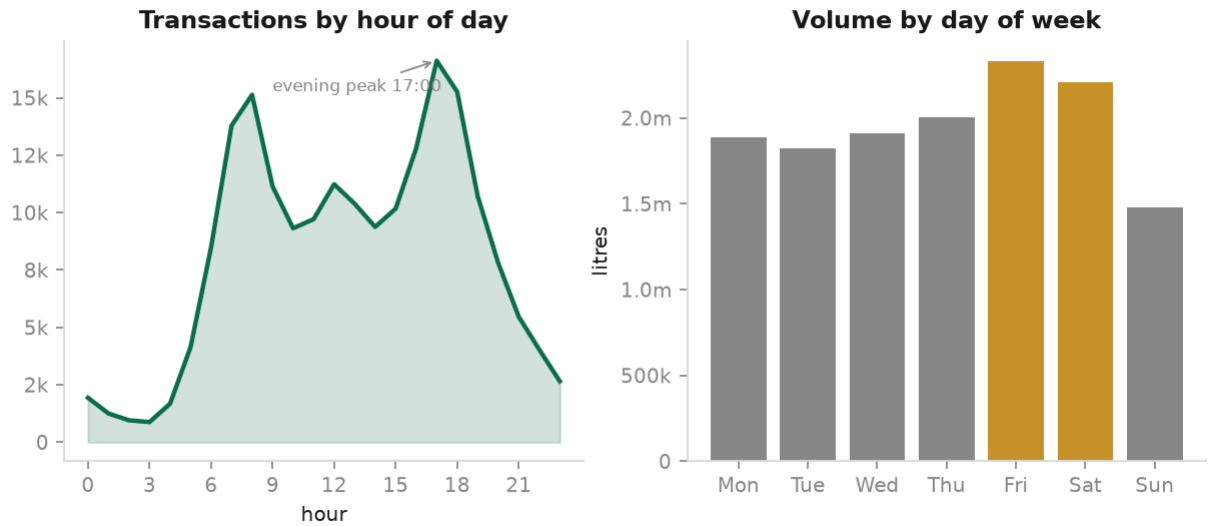
**Capture 1 — not available in this environment.** The Power BI Forecast page before and after a reporting-line filter, showing the score and forecast panels changing together. Native window capture returns a blank image for Power BI Desktop in this session, so no screenshot is presented in its place.

**Capture 2 — not available in this environment.** The Databricks job run history for the medallion pipeline. Native window capture returns a blank image for Power BI Desktop in this session, so no screenshot is presented in its place.

## Demand and trading rhythm

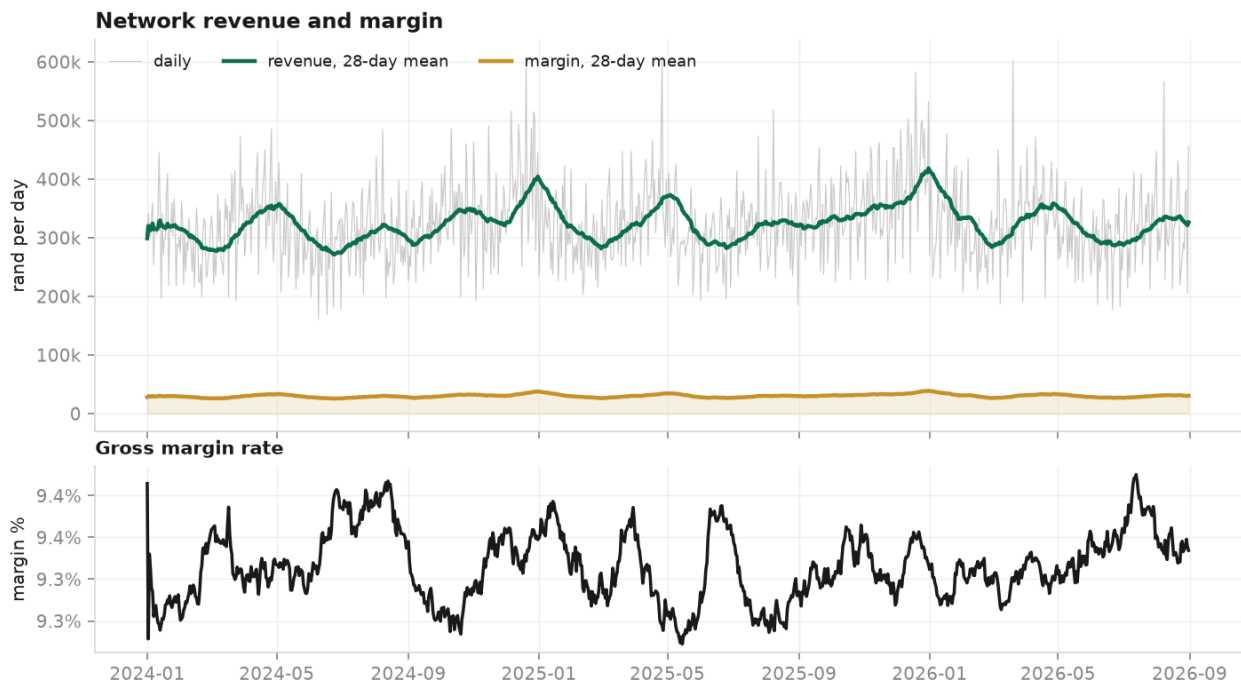
A forecasting model can only learn a pattern the data actually contains. These three figures are the evidence that the generator produces one: an hourly commute shape, a weekly rhythm, and a southern-hemisphere seasonal cycle that survives all the way through to the warehouse.

## Generated demand follows real trading rhythms



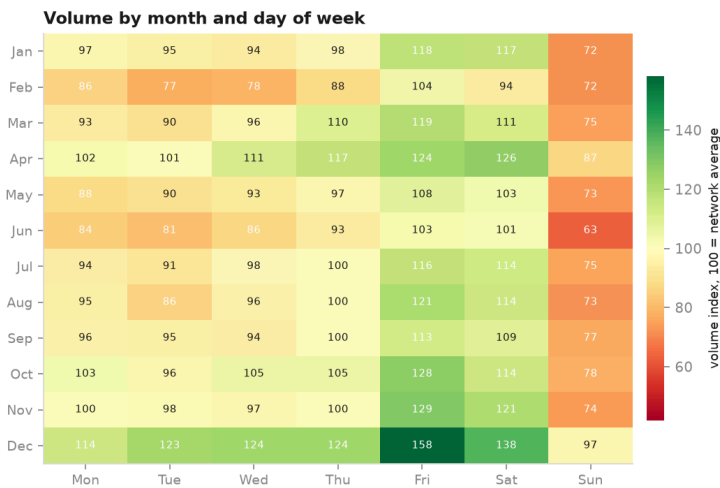
Drawn from drakens360\_test\_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

Commute peaks at 08:00 and 17:00, and a Friday-heavy week.



Drawn from drakens360\_test\_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

Revenue and margin across the modelled window, with the margin rate on its own axis. A rate and a total share an axis only in charts that say nothing about either.



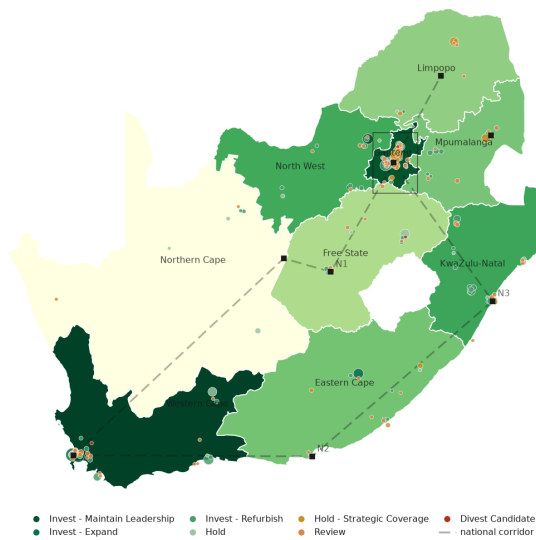
Mean litres per trading day, indexed to 100. Per day rather than per month because the modelled window ends in August: summed, September to December would read a third low and look like seasonality.

Drawn from drakens360\_test\_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

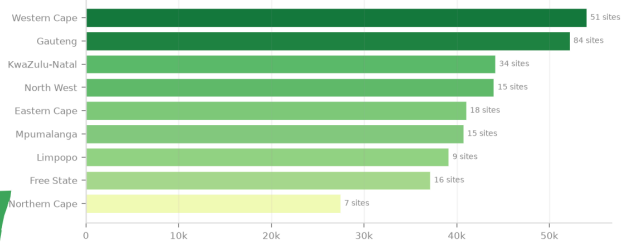
Month against day of week, indexed to the network average. December and January are the high-summer holiday peak; June and July the winter trough.

## Network and investment

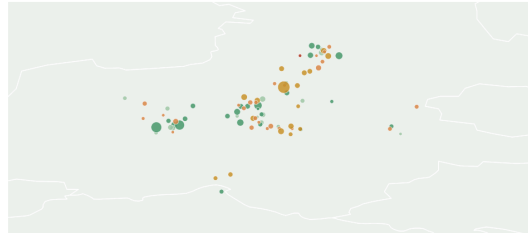
### Recommendation, margin and corridor exposure



### Margin per site, by province



### Gauteng detail (84 sites)



250 synthetic sites across 9 provinces – the South African estate; 250 of 987 sites; the remaining 337 are in other African markets and have no province. Bubble area is total margin; colour is the investment recommendation. Province boundaries are Natural Earth 1:10m admin-1 (public domain); everything drawn on them is generated. Coordinates are South African town centroids plus random jitter and do not represent any real service station. Drawn from drakens360\_test\_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

Provinces shaded by margin per site; bubble area is total margin and colour is the investment recommendation. The N1, N2 and N3 are drawn because corridor sites are protected from divestment regardless of score. Province boundaries are Natural Earth 1:10m admin-1, public domain; everything drawn on them is generated.

### Measure

### Value

Budget

R1.50bn



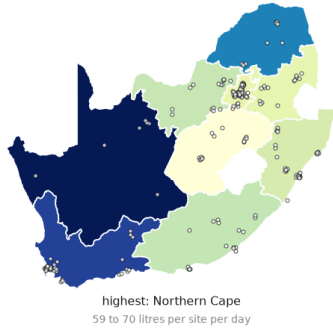
| Measure                | Value          |
|------------------------|----------------|
| Funded projects        | 248            |
| Capital committed      | R1.330bn       |
| Expected annual uplift | R332.9m        |
| Portfolio ROI          | 25.0% per year |
| Blended payback        | 4.0 years      |
| Provinces covered      | 9 of 9         |

## Plan by province

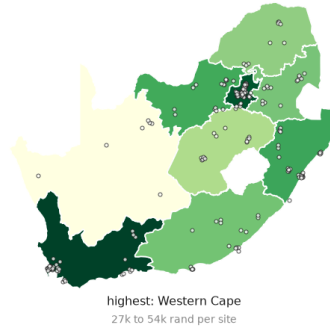
| Province      | Projects | Capex (R m) | Uplift (R m) |
|---------------|----------|-------------|--------------|
| Gauteng       | 84       | 439.20      | 109.11       |
| Western Cape  | 51       | 279.30      | 79.15        |
| KwaZulu-Natal | 34       | 188.00      | 44.93        |
| Eastern Cape  | 16       | 87.30       | 21.48        |
| Free State    | 16       | 85.60       | 19.52        |
| North West    | 15       | 85.00       | 21.03        |
| Mpumalanga    | 15       | 79.30       | 16.77        |
| Northern Cape | 7        | 42.20       | 11.74        |
| Limpopo       | 9        | 38.20       | 8.19         |

## The same network, read six ways

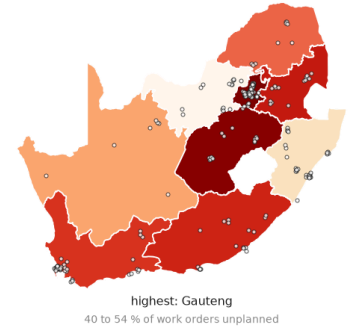
**Throughput**



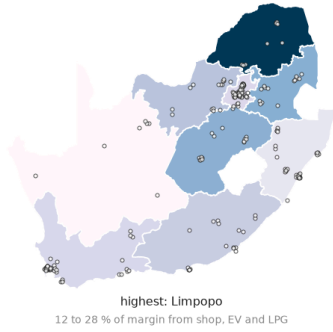
**Margin per site**



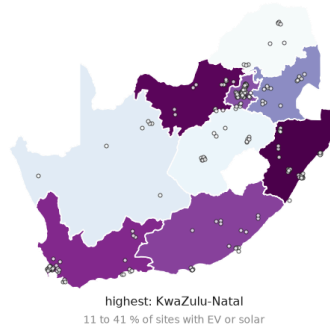
**Asset condition**



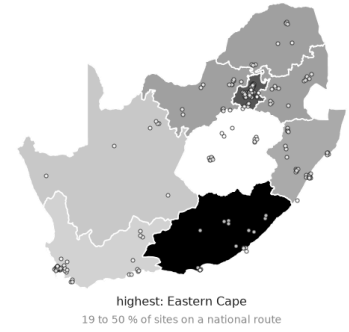
**Non-fuel margin**



**New energy**



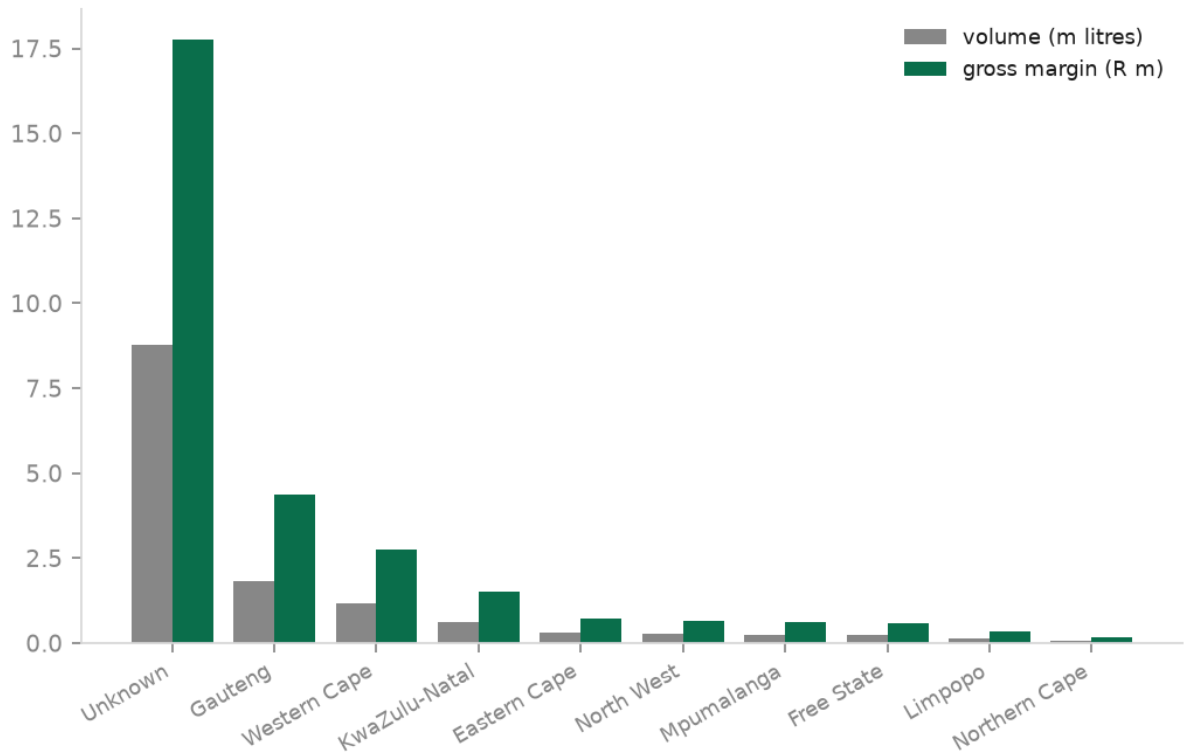
**Corridor exposure**



Provinces shaded by the panel's measure; dots are the synthetic sites, identical in every panel so the six are comparable. Province boundaries are Natural Earth 1:10m admin-1 (public domain). Drawn from drakens360\_test\_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

The same network read six ways, on a fixed basemap and extent so the panels are directly comparable. A single map answers one question; a network planner asks six.

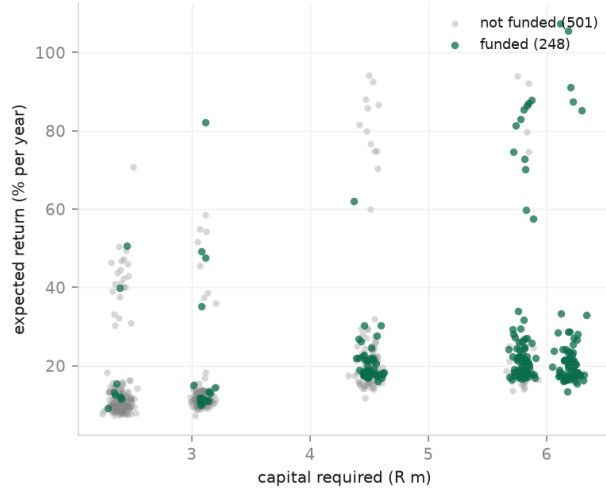
### Volume and margin by province



Drawn from drakens360\_test\_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

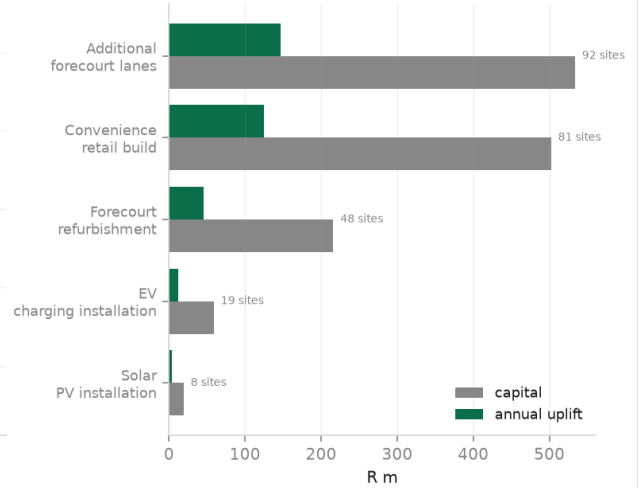
Volume and margin by province.

### Candidate projects: return against capital

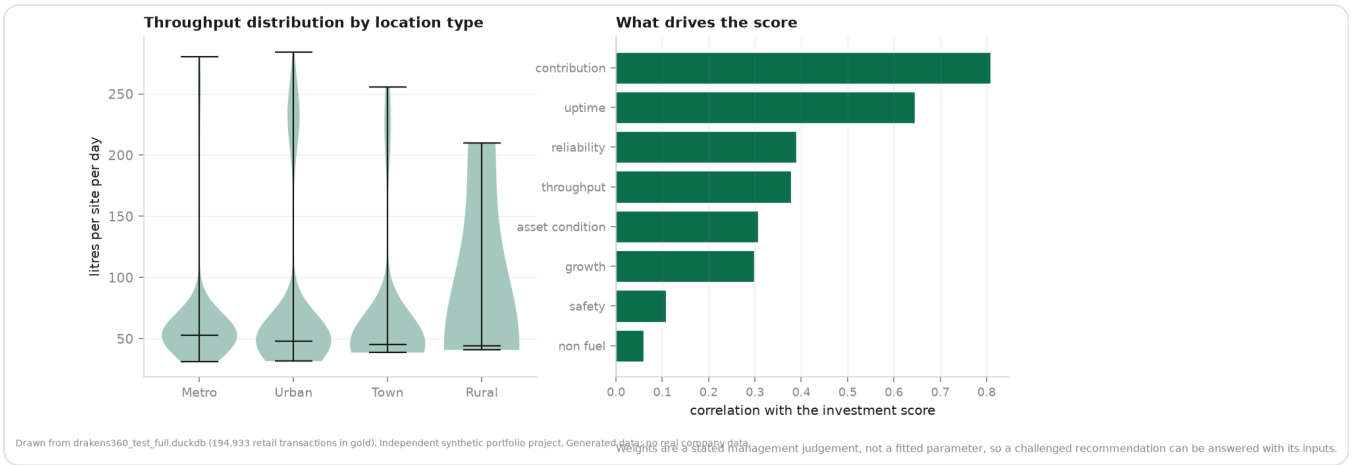


Drawn from drakens360\_test\_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

### Funded plan by intervention



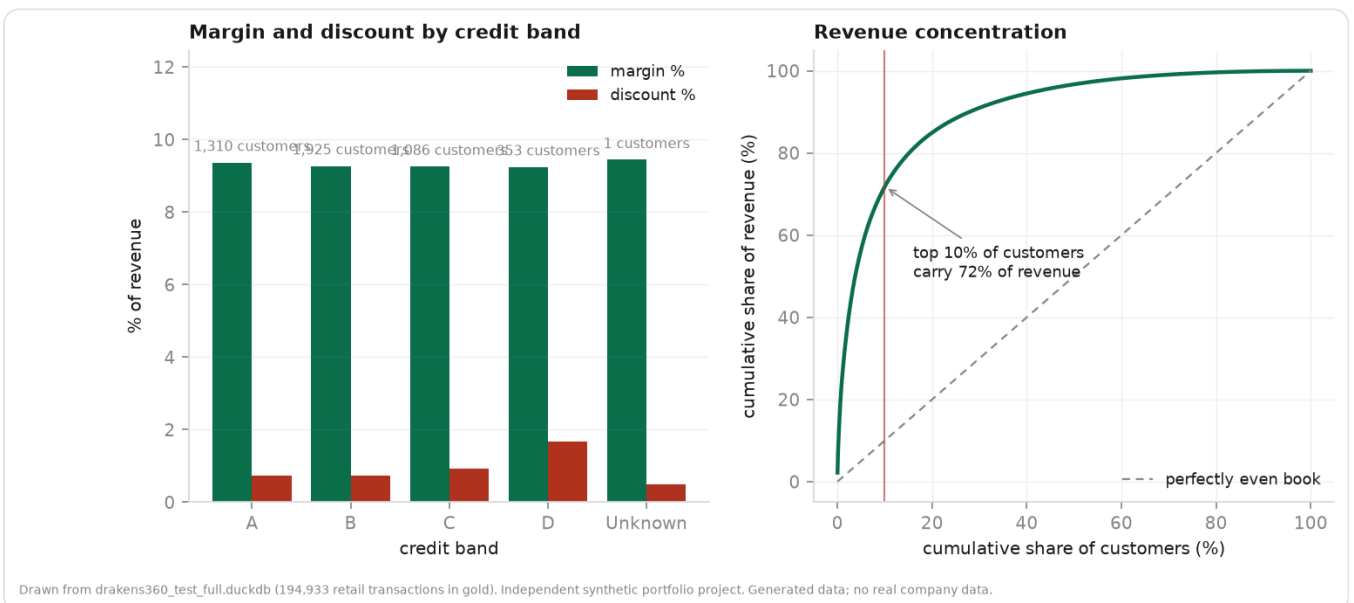
Every candidate project by return against capital, with the funded set highlighted. The funded set is not simply the top of the ROI ranking — provincial minimums and corridor protection pull specific projects in.



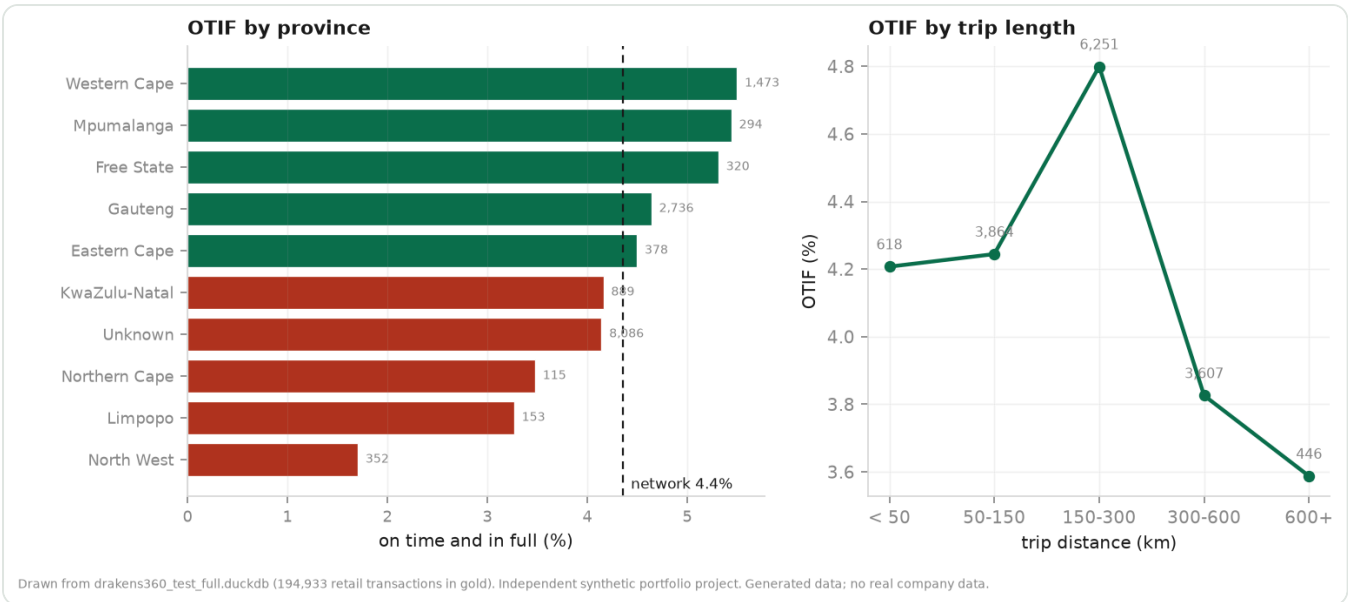
Left: throughput is skewed enough that a network average is a poor summary. Right: what actually drives the investment score, as the correlation between each component and the total.

## Commercial and operations

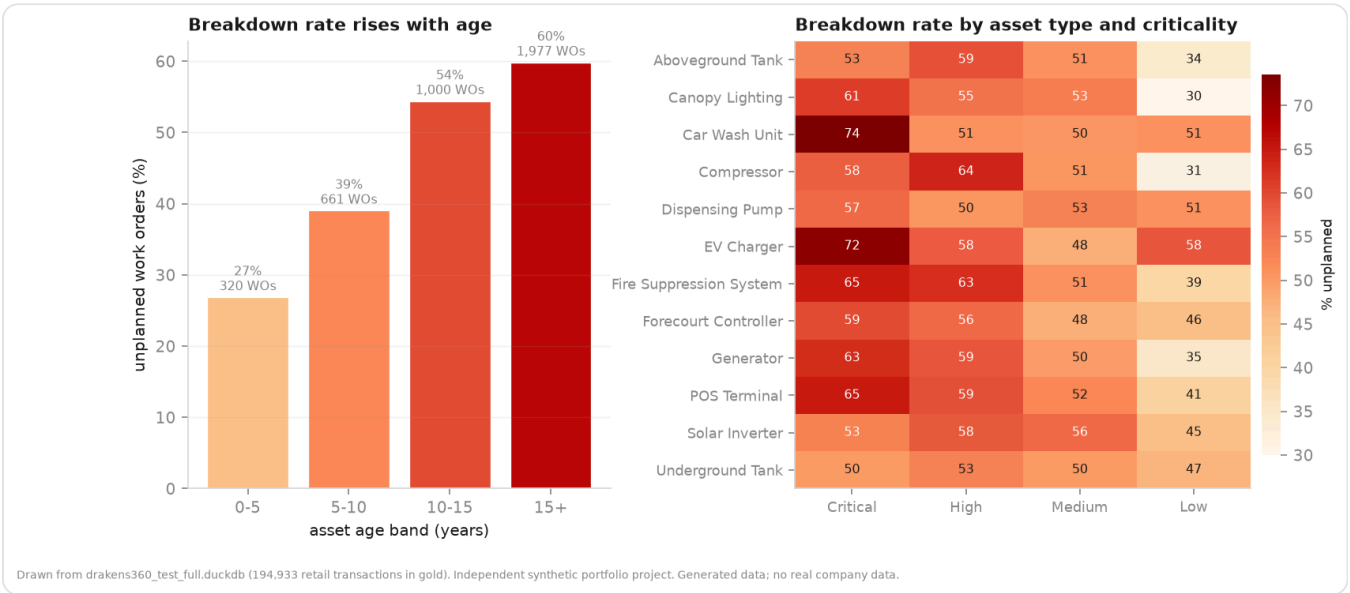
Four questions the business asks that the platform is built to answer, each computed once in gold rather than reassembled per report.



Left: whether the credit band means anything — worse-rated customers discount harder. Right: how concentrated the customer book is, which is a different risk from an even one.



OTIF broken out by province and by trip length. As a single network number it is the least useful form of it: broken out, it says whether lateness is a routing problem, a distance problem or one region's problem.

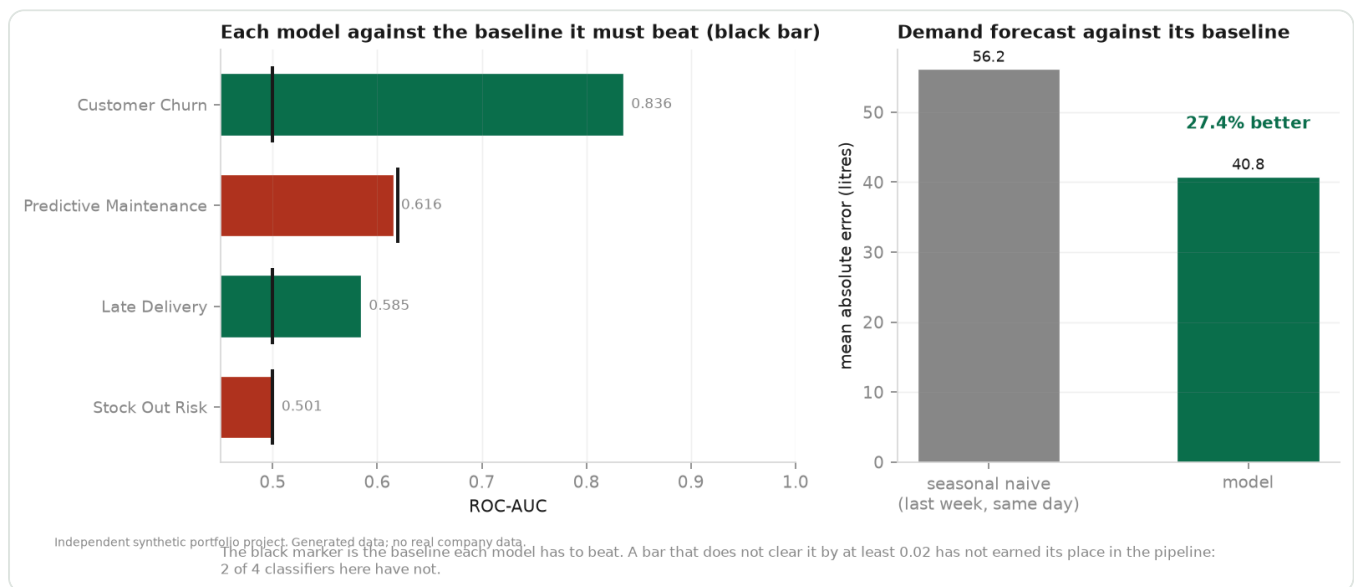


Breakdown rate against asset age and criticality. This is the signal the predictive-maintenance model is allowed to learn from, and the reason that experiment reports an age-only baseline.

## Machine learning

Six experiments, each reported against a baseline it has to beat. Not all of them win, and all six are listed — a portfolio that shows only the models that worked is not showing the job.

| Model                        | Headline result                                                          | Verdict                                             |
|------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------|
| Customer Churn               | ROC-AUC 0.836, 82% precision in the top decile                           | Usable — ranks well above chance                    |
| Demand Forecast              | MAE 40.8 L against a seasonal-naive 56.2 L                               | Beats its baseline by 27.4%                         |
| Fraud Anomaly                | 1.0% of shifts flagged; cash share 30.9% against 23.9% for normal shifts | Flagged shifts differ from normal ones on both axes |
| Late Delivery                | ROC-AUC 0.585, 1.72x lift in the top decile                              | Usable — the reviewed decile is enriched            |
| Predictive Maintenance       | ROC-AUC 0.616, 1.19x lift in the top decile, against a baseline of 0.620 | Does NOT beat its baseline                          |
| Product Demand Forecast      | not run                                                                  | not run                                             |
| Product Revenue Forecast 12M | not run                                                                  | not run                                             |
| Stock Out Risk               | ROC-AUC 0.501                                                            | No usable signal                                    |



Each classifier against the baseline it has to beat, and the demand forecast against its seasonal-naive baseline. A bar that does not clear its black marker has not earned its place in the pipeline.

## What the platform found in itself

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Each of these was caught by the project's own tests, observability or baselines, which is the point of having them.

### **A macro that silently corrupted every ratio**

`safe_divide('a - b', 'c')` compiled to `a - (b / c)` because the macro did not parenthesise its arguments. Not an error — a plausible-looking wrong number. Caught by a range test asserting that `dim_product.base_margin_pct` falls between -5 and 100.

### **A general ledger that did not balance**

Journal lines were posted independently, so debits exceeded credits by 46%. Caught by `obs_reconciliation_controls`, which evaluates double-entry as data rather than only as a test. Fixed by emitting journals as balanced pairs.

### **A partitioning choice that looked obvious and was wrong**

The retail fact was partitioned on `date_key`, because every query filters on a date range. The maintenance job measured the result: 974 files averaging 0.2 MB, which OPTIMIZE can never merge because it does not cross partition boundaries. Under liquid clustering on `(date_key, site_id)` the same 5 million rows occupy 2 files averaging 94 MB — a 487-fold reduction in file count for identical data.

### **A leak that scored 0.998**

The stock-out model was scored against the current inventory snapshot. Its target is `days_of_cover` below a threshold; `days_of_cover` is stock over usage; and both components were features. Excluding `days_of_cover`, which the code did with a comment explaining why, excluded nothing — the model divided one component by the other and reproduced the rule. Re-pointed at the next snapshot it scores 0.510.

### **A model with no baseline, losing to one variable**

Predictive maintenance reported a 1.35x lift and passed review. It had no baseline. Ranking the queue by asset age alone scores ROC-AUC 0.620 against the model's 0.598, and the reported lift moves between 1.03 and 1.35 across capacity settings with no trend. Every experiment now computes and logs the baseline it has to beat.

### **Null foreign keys where an unknown member belonged**

The landing zone contains referential drift by design — a site code in a fact that the dimension never received. Gold left-joined those facts and kept the null key, which drops the row from every inner join and from any total grouped by province: 5,070 retail transactions with real litres and real margin were invisible to regional reporting. They now resolve to a -1 unknown member, so nothing disappears from a total and the drift is countable.

### **A rule inferred from a column name, four times over**

The pattern recurred often enough to be worth naming. A validity rule written against what a column is *called* rather than what it *means* destroys real data, and does it silently while every model builds and every test passes. ``signed_amount_zar >= 0`` quarantined every credit leg of the general ledger. A 2023-2027 timestamp window nulled 89% of site opening dates. An 8-45 rand fuel price band rejected 36% of the shop feed, where the same column name carries the price of a pie. Requiring a site on every row quarantined 58% of digital events, which are app sessions and were never at a station. Bands are now keyed by table and column, attribute dates get a window wide enough for a business history, and optional foreign keys are measured rather than assumed.

### **A tab that split a dimension in two**

The maintenance heatmap showed thirteen asset types where the dimension has twelve. `clean_text` called `trim()`, which in both DuckDB and Spark removes spaces and nothing else, so a tab-padded value reached gold as a separate category from the same value without one. Both real, both counted separately, neither obviously wrong in a total.

### **Observability that measured 8% of what it claimed**

`obs_cleansing_summary` named fourteen tables against a landing zone of 167, and the fourteen were whichever tables someone had thought to add — the least useful sample available. It is now generated by the same pass that generates the cleansing models: 128 monitored tables, 2,049,277 raw rows reconciling exactly to cleansed plus quarantined plus deduplicated, and zero tables failing reconciliation.

### **A validity rule that deleted half the general ledger**

The contract applied a blanket rule to every amount column: anything named `*_zar` must be non-negative. `signed_amount_zar` is negative on the credit side by design, so the rule quarantined 41,886 of the ledger's 84,000 lines — every credit leg of every journal. What survived was a ledger of debits with almost no credits, which reported a 99.98% imbalance that looked like a generator bug and was a contract bug. A validity rule inferred from a column name is a guess, and a wrong guess deletes data silently instead of failing loudly. Signed measures are now matched by naming convention and excluded, so a new `net_` or `_variance` column is covered the day it is added.



### **Double-entry integrity is a property of the journal, not the line**

Cleansing assesses rows one at a time, so when one leg of a balanced pair failed its contract the other leg survived alone. Each orphaned leg sat in the ledger as an unmatched debit or credit, and the ledger could not balance however correct the generator was — the imbalance was created by the cleansing step itself. The mart now publishes only journals whose lines still net to zero, and the orphans stay in quarantine beside the leg that was rejected. Debits and credits now agree exactly, to the cent.

### **A sentinel that passed every non-negativity check**

The generator uses 0 as one of its numeric sentinels, and 0 satisfies every 'amount  $\geq 0$ ' rule. Stripping zero the way -999 is stripped is not an option, because zero litres and zero margin are legitimate readings. The contract now checks amounts against their own row instead: a transaction of 82 litres at R22.34 reporting R0.00 of sales is internally inconsistent whatever caused it.

## **Limitations**

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Stated plainly, because a report that claims completeness is misleading.

- The Databricks results are at portfolio scale (5,000,000 retail rows through bronze, silver and gold). The DuckDB warehouse figures in this report are at test scale, and each figure is stamped with the warehouse it came from. The two should not be read as one run.
- Predictive maintenance does not beat an age-only baseline, and stock-out risk has no signal at the current snapshot cadence. Both limits are in the generated data rather than in the estimators, and both fixes are changes to the generator: condition monitoring for the first, denser inventory readings for the second.
- The investment plan's rand figures rest on one stated benchmark — what a mid-sized site earns in fuel gross margin in a year — because the transaction fact is a sample and capital costs are not. Rank order and every ratio between sites are unaffected; only the level is rebased.
- Row filters are correct but their query cost under concurrent executive load has not been measured.
- There is no data subject access or erasure workflow. The masking and pseudonym design supports one; the process itself is not built.

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Drakens Energy is a fictional company invented for this project. It does not exist. Every site, customer, employee, supplier, asset, coordinate, price, volume, margin and incident in this report is randomly generated. Site coordinates are South African town centroids plus random jitter and do not represent any real service station.